Assignment 4

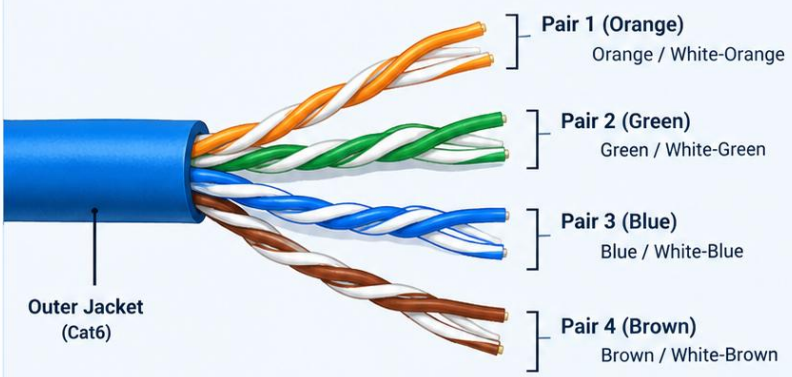
Network Cables & Connectors

1. **Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.**

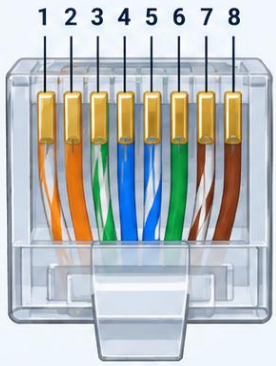
Cat6 Ethernet Cable – T568B Standard

(Color order + RJ-45 connector pinout)

1. Inside the Cat6 Cable (T568B Color Order)



2. RJ-45 Connector (Front View)

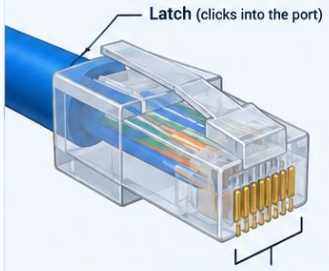


RJ-45 Plug
(looking at the pins from the front)

3. T568B Color Order (Pinout)

Pin No.	Wire Color (T568B)	Pair
1	White-Orange	Pair 2 (Orange)
2	Orange	Pair 2 (Orange)
3	White-Green	Pair 3 (Green)
4	Blue	Pair 1 (Blue)
5	White-Blue	Pair 1 (Blue)
6	Green	Pair 3 (Green)
7	White-Brown	Pair 4 (Brown)
8	Brown	Pair 4 (Brown)

4. RJ-45 Connector (Side View)



5. Quick Notes

- **T568B is the most** common standard (used in home networks and many offices).
- Keep the same color **order** on both ends of the cable (straight-through cable).
- For a **crossover cable**, one end is T568A and the other end is T568B.
- **Cat6 supports** up to **10 Gbps** (up to 55 meters) and 1 Gbps (up to 100 meters).

T568B = Orange-White, Orange, Green-White, Blue, Blue-White, Green, Brown-White, Brown
(Pins 1 → 8)

2. **List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.**

Network Cable	**Real-world use**	**Why?**
Cat5e	Wi-Fi Router	Affordable and supports up to **1 Gbps**, enough for **normal home/office internet**.
Cat6	Zomato POS Terminal	Provides **faster and more reliable wired communication** between the POS system and network.
Fiber Optic	IPL Streaming Server	Supports very **high-speed data over long distances** with low signal loss, suitable for streaming large amounts of video data.

3. **Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.**

Steps to Crimp an RJ-45 Connector

1. Cut the cable to the required length.
2. Remove the outer jacket from about 1 inch of the cable.
3. Separate and straighten the 8 wires inside the cable.
4. Arrange the wires in T568B order:
   White-Orange → Orange → White-Green → Blue → White-Blue → Green → White-Brown → Brown.
5. Cut the wires evenly so all 8 ends are the same length.
6. Insert the wires into the RJ-45 connector, making sure they reach the end and stay in the correct order.
7. Place the connector in the crimping tool and press it firmly to lock the wires in place.
8. Test the cable with a LAN cable tester to make sure all pins work correctly.

Two Common Mistakes to Avoid

- Wrong wire order: Mixing up the T568B color sequence can make the cable fail.
- Loose or short wires: If the wires do not reach the end of the connector properly, the pins may not make good contact.

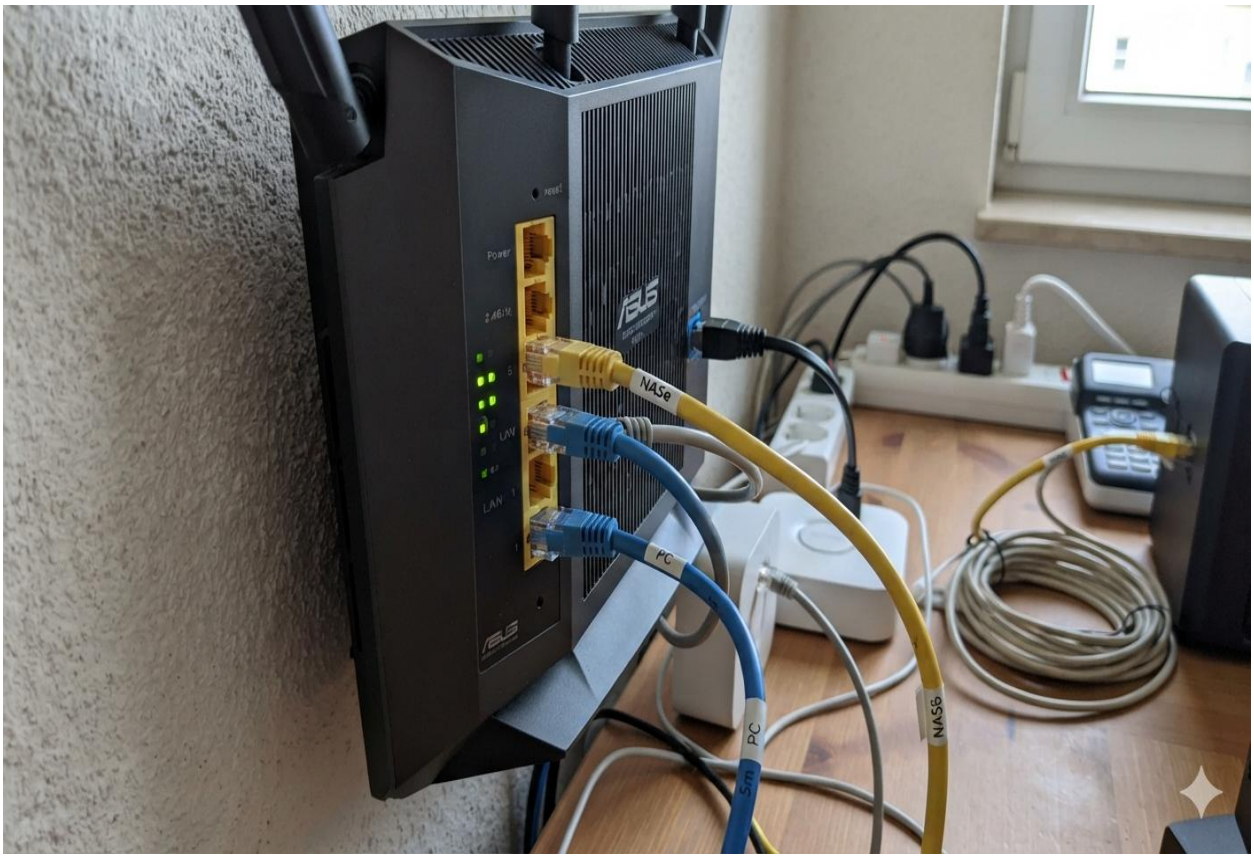
4. **Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.**

For an IPL screening gaming zone, I would use:

Connection	Cable	Connector	Reason
Console → TV/Monitor	HDMI 2.1	HDMI connector	Supports high resolution, high refresh rates, and carries both video and audio
TV/Monitor → Audio system	HDMI ARC/eARC	HDMI connector	Sends high-quality audio to speakers/soundbar
Console/PC → Network switch	Cat6 Ethernet	RJ-45	Fast and reliable network connection with low latency
Switch → Router	Cat6 Ethernet	RJ-45	Provides stable internet for multiplayer gaming

- **HDMI 2.1 is suitable for connecting consoles to screens because it provides high-speed video and audio. Cat6 with RJ-45 is preferred for gaming network connections because it gives a stable, low-latency connection compared with Wi-Fi. This setup helps keep both the IPL screening and gaming experience smooth and reliable.**

5. **Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.**



This photo shows a home network setup with a Wi-Fi router connected to different devices using Ethernet cables with RJ-45 connectors.

- **Blue Ethernet cables:** Connect the router to a PC and Smart Home Hub.
- **Yellow Ethernet cable**: Connects the router to a NAS (Network-Attached Storage) device.
- **WAN cable:** Connects the router to the modem/internet connection.
- These are twisted-pair Ethernet cables, likely **Cat5e** or **Cat6**.
- The RJ-45 connectors are plugged into the router's LAN/WAN ports to provide wired network connectivity**.**